

1(a)	any two from: 1 mass of (mung bean) <u>seedlings</u> ; 2 age / variety / type / cultivar (of mung bean) <u>seedlings</u> ; 3 grinding time (using pestle and mortar) ; 4 speed of centrifuge / time in centrifuge ;	2
1(b)(i)	<i>explanation</i> <i>idea of</i> the temperature (of, solutions / reactants) will not have reached 30 °C / AW or the rate of (enzyme)reaction, will change / not be constant / AW or reaction will start before reaching <u>30 °C / AW</u> ; <i>modification</i> equilibrate / AW, phosphatase/enzyme/extract and PPP/substrate, separately or equilibrate / AW , phosphatase / enzyme / extract and PPP / substrate before mixing ;	2
1(b)(ii)	(sodium carbonate solution) denatures, the enzyme / phosphatase ;	1

Question	Answer	Marks																																		
1(c)	1 five stated concentrations between 2 % and 0 % and units, % / percentage ; 2 correct method for dilution shown for two <u>intermediates</u> ; <table border="1"> <thead> <tr> <th rowspan="2">concentrations %</th><th>2 % stock /</th><th rowspan="2"><u>distilled</u> water / cm³</th></tr> <tr> <th>phenolphthalein / cm³</th></tr> </thead> <tbody> <tr><td>2.0</td><td>50</td><td>0</td></tr> <tr><td>1.6</td><td>40</td><td>10</td></tr> <tr><td>1.5</td><td>37.5</td><td>12.5</td></tr> <tr><td>1.2</td><td>30</td><td>20</td></tr> <tr><td>1.0</td><td>25</td><td>25</td></tr> <tr><td>0.8</td><td>20</td><td>30</td></tr> <tr><td>0.5</td><td>12.5</td><td>37.5</td></tr> <tr><td>0.4</td><td>10</td><td>40</td></tr> <tr><td>0.25</td><td>6.25</td><td>43.75</td></tr> <tr><td>0.0</td><td>0</td><td>50</td></tr> </tbody> </table>	concentrations %	2 % stock /	<u>distilled</u> water / cm ³	phenolphthalein / cm ³	2.0	50	0	1.6	40	10	1.5	37.5	12.5	1.2	30	20	1.0	25	25	0.8	20	30	0.5	12.5	37.5	0.4	10	40	0.25	6.25	43.75	0.0	0	50	2
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1(d)	1 absorbance / AW, for the dilutions <u>and</u> extract / reaction mixture ; 2 use the calibration, curve / graph (to determine phenolphthalein / extract concentration) ;	2																																		

Question	Answer	Marks
1(e)	any six from: 1 prepare five solutions with stated pH values up to pH 7 ; 2 equilibrate / AW, PPP / substrate, <u>and</u>, phosphatase / enzyme / extract, <u>and</u> buffer solutions, separately / before mixing ; 3 mix buffered, PPP / substrate, with, extract / enzyme solution ; 4 calibrate colorimeter or use a blank / AW, to, set colorimeter (absorbance) to zero / reset colorimeter ; 5 at each pH, stop the reaction / AW after a set time <u>and</u>, measure / note / record, absorbance ; 6 suitable method to identify the optimum pH ; 7 repeat experiment with smaller pH intervals around, the optimum pH / values with the highest absorbance / concentration (of phenolphthalein) ; 8 repeat experiment at least twice and calculate the mean (absorbance), for each pH / their means ; 9 named hazard <u>and</u> risk <u>and</u> precaution ;	6

Question	Answer			Marks
1(e)	hazard	risk	precaution	
	phosphatase enzyme / mung bean / extract	allergy / irritant	gloves / eye protection / PPE	
	phenolphthalein phosphate (PPP)			
	sodium carbonate			
	pH buffer	irritant pH 3–7 or corrosive below pH 4	gloves / eye protection / PPE	
	phenolphthalein	toxic / irritant / allergy	gloves / eye protection / PPE	
		flammable	keep away from naked flame	
1(f)(i)	point 6 on top line is circled ; higher than, the $V_{\text{max}} / 0.018 \text{ (mmol dm}^{-3} \text{ s}^{-1}) / \text{the plateau ;}$			2
1(f)(ii)	(K _m with no inhibitor) 0.075 <u>and</u> (K _m with inhibitor) 0.43 / 0.425 <u>and</u> mmol dm ⁻³ ;			1
1(f)(iii)	1 <i>working</i> difference divided by K _m with no inhibitor (×100) (0.425 – 0.075) / 0.075 (×100) ; 2 correct answer from working in mp1 ;			2